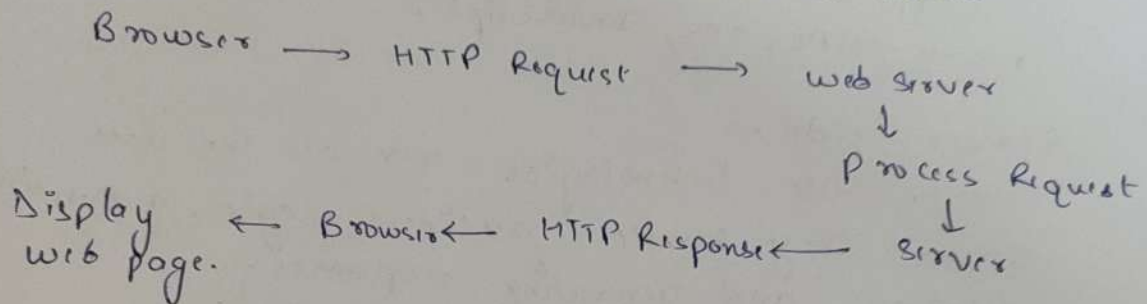


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Frontend Assignment → 1

1) The HTTP Request - Response cycle is the process in which a browser communicates with a web server to request and receive a resource.



2) HTML → HTML (Hypertext Markup Language) is used to create the structure and content of a webpage.

CSS → CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. It manages colors, fonts and spacing.

JavaScript → JavaScript is a programming language used to add interactivity and dynamic behaviour to webpages.

Bootstrap → Bootstrap is a front-end framework used to create responsive and mobile friendly websites.

JQuery → JQuery is a Javascript library that simplifies DOM manipulation, event handling

React.js → React.js is a javascript library for creating interactive user interfaces. It uses reusable components and virtual DOM for efficient updates.

3) client-side technologies run mainly in the user's browser. They handle the webpage's interface and user interaction.

Ex — HTML, CSS, JavaScript.

Server-side →

Server side technologies run on the web server. They handle business logic, authentication, databases, and generating responses.

Ex — Node.js, Python, Java, PHP

4) Block-level elements normally start on a new line and occupy the available width of their parent container.

Characteristics →

- Start on a new line
- Usually occupy the full available width
- Used to organize webpage content.

Ex — <div>, <section>, <header>

5) Semantic HTML elements clearly describe the meaning and purpose of their content.

Importance:

- Improves code readability.
- Improves accessibility
- Helps search engines understand the webpages
- Makes code easier to maintain.

Examples →

<header> , <footer> , <nav> , <main> , <section>

unlike general div semantic provide meaningful information.

6)

```
<table border = "1">
```

```
  <tr>
```

```
    <th> Sr. No </th>
```

```
    <th> Name </th>
```

```
    <th> Roll No </th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td> 1 </td>
```

```
    <td> John </td>
```

```
    <td> 34 </td>
```

```
  </tr>
```

```
  <tr>
```

```
    <td> 2 </td>
```

```
    <td> Smith </td>
```

```
    <td> 43 </td>
```

```
  </tr>
```

```
</table>
```

```

7)
<ul>
  <li> River
    <ul>
      <li> Ganga </li>
      <li> Yamuna </li>
      <li> Narmada </li>
    </ul>
  </li>
</ul>

```

The outer creates the main list, while the inner creates the nested list.

8) There are three types of CSS

1. Inline CSS

CSS is written directly inside an HTML element

```
<p style="color: blue"> okay </p>
```

2. Internal CSS

This is written inside a style tag in an HTML document.

3. External CSS

This CSS is written in a separate .css file and linked to HTML

```
<link rel="stylesheet" href="style.css">
```

9) HTML, CSS, and JavaScript have different roles:

- HTML → structure : creates webpage, content and elements.
- CSS → Presentation : controls colors, layout, fonts and appearance.
- JavaScript → Behaviour : Adds interaction and dynamic functionality.

Thus, HTML defines what is present, CSS defines how it looks and JavaScript defines how it behaves.

10) a) Role of Bootstrap —

Bootstrap is a front-end framework used to create responsive and mobile-friendly websites quickly. It provides grid system and ready-made components.

(b) JQuery → JQuery is a JavaScript library that simplifies DOM manipulation, event handling, animations and AJAX. It was widely because it made JS development easier.

(c) React.js is a JavaScript library used to build dynamic and interactive user interfaces. It uses reusable component and virtual DOM to efficiently update webpage.